Materials for Energy (ChE 494)

(Spring 2020; Department of Chemical Engineering, University of Illinois at Chicago)

Instructor: Sanjay K. Behura (Office: EIB RM142; E-mail: sbehura1@uic.edu; Contact: 312-355-2914)

Class Schedule: When: Tuesday and Thursday, 8:00 AM - 9:15 AM; **Where:** EIB 124

Course Objectives: The goal of this course is to prepare students with foundational understanding of the advanced materials including low-dimensional nanostructures, tools required to synthesize and characterize such materials, and their potential applications in photovoltaic energy conversion.

Detailed Course Contents:

Science of Advanced Materials (3 Weeks): Quantum Theory of Solids; Semiconductor Physics; Low-Dimensional Materials Science; Modifications of Materials Properties. (This part will focus on physics and chemistry of materials, their unique structure and intriguing properties).

Advanced Materials Processing (2 Weeks): Synthesis Techniques for Advanced Material Thin Films: Physical Vapor Deposition and Chemical Vapor Deposition. (This part will focus on various strategies for the nucleation of materials and understanding the ensuing growth mechanisms).

Material Characterization (3 Weeks): Analysis of Materials by Raman Spectroscopy, X-Ray Photoelectron Spectroscopy, Scanning Electron Microscopy, Transmission Electron Microscopy, Scanning Probe Microscopy, and Electrical Characterization. (This part will focus on basic working principles of advanced characterization tools).

Principles of Solar Energy Harvesting (5 Weeks): Properties of Sunlight; Solar Cell Operation; Design and Development of Advanced Solar Cells: Silicon Homojunction and Graphene-Based Schottky-Junction Solar Cells, Organic and Dye-Sensitized Molecular Solar Cells, and Hybrid Organic-Inorganic Perovskite Photovoltaic Cells.

Energy Lab Work (2 Weeks): (i) Synthesis and Characterization of Low-Dimensional Materials (2D-Graphene and 1D-ZnO) and (ii) Design and Development of Dye-Sensitized Molecular Cells.

Grading Policy: Quiz/Exam: 40%; Homework Assignments: 10%; Mid-term Article Writing: 25%; End-term Project Presentations: 25%.

Textbooks:

(1) Introduction to Solid State Physics (Charles Kittel, 8th Edition, Wiley)

(2) 2D Materials: Properties and Devices (P. Avouris, T. F. Heinz, and T. Low)

(3) Physics of Solar Cells (Peter Wurfel)

(4) Semiconductor Material and Device Characterization (Dieter K. Schroder, 3rd Edition, Wiley)